**Question 1.** 1. Proposition 3.1b in Humphreys states if $I$ is a solveable ideal of $L$ such that $L/I$ is solveable, then $L$ is solveable. Since $L/[L,L]$ is abelian by definition of $[L,L]$ and $[L,L]$ is nilpotent hence solveable we conclude by the proposition that $L$ is solveable. □

2. We first note that $\mathrm{ad} : L \to \mathfrak{gl}(L)$ is a homomorphism, to see this we employ the Jacobi identity

$$[\mathrm{ad}_x, \mathrm{ad}_y](z) = \mathrm{ad}_x \mathrm{ad}_y(z) - \mathrm{ad}_y \mathrm{ad}_x(z) = [x,[y,z]] - [y,[x,z]] \overset{\text{Jacobi}}{=} [[x,y],z] = \mathrm{ad}_{[x,y]} \tag{1}$$

Thus by Humphreys Proposition 3.1(a) $L$ being solveable implies $\mathrm{ad}(L)$ is a solveable subalgebra of $\mathfrak{gl}(L)$. Thus by Lie's theorem there exists a basis of $L$ such that $\mathrm{ad}(L) \subset \mathfrak{t}(F, \dim L)$ under the identification of $\mathfrak{gl}(L)$ with $\mathfrak{gl}(F, \dim L)$. In part 4 of this problem we show that $[\mathfrak{t},\mathfrak{t}] = \mathfrak{n}$ the subalgebra of strictly upper triangular matrices. Thus it follows that

$$\mathrm{ad}[L,L] = [\mathrm{ad}\, L, \mathrm{ad}\, L] \subset [\mathfrak{t},\mathfrak{t}] = \mathfrak{n}(F, \dim L) \tag{2}$$

Since elements of $\mathfrak{n}(F, \dim L)$ are nilpotent, equation (2) tells us that $[L,L]$ is ad-nilpotent, thus by Engel's theorem $[L,L]$ is nilpotent. □

3. Consider the nonabelian 2 dimensional Lie algebra $L = \mathrm{span}_F(x,y)$ with relation $[x,y] = x$.

$$L^{(1)} = \mathrm{span}_F \{x\}, \quad L^{(2)} \overset{\text{def}}{=} [\mathrm{span}_F \{x\}, \mathrm{span}_F \{x\}] = 0 \tag{3}$$

However the central series has $L^1 = \mathrm{span}_F \{x\}$. It follows by induction that $L^k = [L, L^{k-1}] = [L, \mathrm{span}_F \{x\}] = \mathrm{span}_F(x)$ for all $k > 1$, hence $L$ is not nilpotent. If you don't believe me that $L$ is a lie algebra I give a calculation of the Jacobi identity.

$$[x,[x,y]] + [x,[y,x]] + [y,[x,x]] = [x,x] + [x,-x] + [y,0] = 0 + -[x,x] + 0 = 0 \tag{4}$$
$$[y,[x,y]] + [x,[y,y]] + [y,[y,x]] = [y,x] + [x,0] + [y,-x] = [y,x] + 0 - [y,x] = 0 \tag{5}$$

□

4. Throughout this exercise I will denote $\mathfrak{n}$ as the subset of strictly upper triangular matrices, and for $k \geq 1$ I will denote $\mathfrak{n}_k$ as the subset of matrices $\{\sum a_{ij} e_{ij} \mid a_{ij} \in F, j \geq i+k\}$. I will first compute the lower central series:

*Claim 1.* $[L,L] \subset \mathfrak{n}$

*Claim 2.* $[L,\mathfrak{n}] \supset \mathfrak{n}$

Claim 2 tells us in particular that $[L,L] \supset [L,\mathfrak{n}] \supset \mathfrak{n}$, so that using claim 1, $[L,L] = \mathfrak{n}$, it follows by induction and claim 1 that that $L^k = [L, L^{k-1}] = [L,\mathfrak{n}] = \mathfrak{n}$ is the lower central series.

The derived series requires sharper claims:

*Claim 3.* $[\mathfrak{n}_k, \mathfrak{n}_r] \subset \mathfrak{n}_{k+r}$

*Claim 4.* $[\mathfrak{n}_k, \mathfrak{n}_r] \supset \mathfrak{n}_{k+r}$

Using claims 3 and 4 and the solution for the lower central series we get the following derived series (note that for $k \geq n$ we have $\mathfrak{n}_k = 0$) $L^0 = \mathfrak{t}$, $L^1 = L^1 = \mathfrak{n}$ then by induction and using claims 3 and 4, we find for $k > 1$ that $L^{(k)} = [\mathfrak{n}_{2^{k-2}}, \mathfrak{n}_{2^{k-2}}] = \mathfrak{n}_{2^{k-1}}$. □

*Proof of Claim 1.* Let $A = \sum a_{ij} e_{ij}, B = \sum b_{ij} e_{ij} \in L$ and let $C = [A,B] = \sum c_{ij} e_{ij}$, then it suffices to show that $c_{jj} = 0$ for each $j$, this is straightforward since

$$c_{jj} = \sum_k a_{jk} b_{kj} - b_{jk} a_{kj} \tag{6}$$

but then since $a_{jk} = 0 = b_{jk}$ for $k < j$ and $b_{kj} = a_{kj} = 0$ for $k > j$ we find that

$$c_{jj} = a_{jj} b_{jj} - b_{jj} a_{jj} = 0 \tag{7}$$

□

*Proof of Claim 2.* It suffices to show that $e_{ij} \in [L, \mathfrak{n}]$ for all $j > i$, so let $j > i$ then $e_{ii} \in L$ and $e_{ij} \in \mathfrak{n}$ so that

$$[e_i i, e_{ij}] = e_{ii}e_{ij} - e_{ij}e_{ii} = e_{ij} \in [L, \mathfrak{n}] \tag{8}$$

□

*Proof of Claim 3.* Let $A = \sum a_{ij}e_{ij} \in \mathfrak{n}_s$ and $B = \sum a_{ij}e_{ij} \in \mathfrak{n}_r$, it suffices to show that $AB \in \mathfrak{n}_{s+r}$, because then by symmetry te same result will hold for $BA$ and thus also their difference. So let $AB = \sum c_{ij}e_{ij}$, it follows that

$$c_{ij} = \sum_k a_{ik}b_{kj} \overset{A\in\mathfrak{n}_s}{=} \sum_{k\geq i+s} a_{ik}b_{kj} \overset{B\in\mathfrak{n}_r}{=} \sum_{j-r\geq k\geq i+s} a_{ik}b_{kj} \tag{9}$$

Thus the sum is only nonzero when $j \geq i + s + r$ thus $c_{ij} = 0$ for all $j < i + s + r$, this proves that $AB \in \mathfrak{n}_{s+r}$. □

*Proof of Claim 4.* It suffices to show that $e_{ij} \in [\mathfrak{n}_s, \mathfrak{n}_r]$ whenever $n \geq j \geq i + s + r$. Choose such an $e_{ij}$, then $e_{i,j-r} \in \mathfrak{n}_s$ and $e_{j-r,j} \in \mathfrak{n}_r$ so that

$$[e_{i,j-r}, e_{j-r,j}] = e_{i,j-r}e_{j-r,j} - \underbrace{e_{j-r,j}e_{i,j-r}}_{=0 \text{ since } j>i} = e_{ij} \in [\mathfrak{n}_s, \mathfrak{n}_r] \tag{10}$$

□